IBP for Bubble

Opencode

Abstract

In this note we construct IBP relations for bubble-like cosmological correlators.

Contents

1 Massless Case in de Sitter Space **2**
- 1.1 Feynman Rules for Massless Scalar Field . . . 3
- 1.2 One-Loop Two-Point Function for Massless Case . . . 4
- 1.3 Integral Family for IBP Analysis . . . 4
- 1.4 IBP Relations . . . 5
- 1.5 Momentum IBP Relations . . . 5
- 1.6 Momentum IBP Relations . . . 5
- 1.7 Symmetry Relations Under Propagator Exchange . . . 6
- 1.8 Complete IBP System . . . 7
  - 1.8.1 Time IBP Relations . . . 7
  - 1.8.2 Momentum IBP Relations . . . 7
  - 1.8.3 Symmetry Relations . . . 8

2 Vertex++ IBP Relations **9**
- 2.1 Definition of Vertex++ Integrals . . . 9
- 2.2 Boundary Terms from Time Derivatives . . . 10
- 2.3 Reduction of Boundary Integrals H . . . 11
  - 2.3.1 Time IBP for H . . . 11
  - 2.3.2 Momentum IBP for H . . . 11
  - 2.3.3 Symmetry Relations . . . 11
  - 2.3.4 Reduction Strategy . . . 12
- 2.4 Momentum IBP Relations . . . 12
- 2.5 Algebraic Relations and Symmetries . . . 12
- 2.6 Complete IBP System for V_{++} . . . 13
- 2.7 Relation to Chapter 1 IBP and Separate Reduction Strategy . . . 13
  - 2.7.1 Why Separate Reduction is Preferred . . . 14
  - 2.7.2 Energy Relabeling as a Simplification Tool . . . 14
  - 2.7.3 Consequences for the Reduction Procedure . . . 14
- 2.8 Practical Reduction Strategy . . . 15
  - 2.8.1 Reduction Procedure for Separate Time-Orderings . . . 15
  - 2.8.2 Key Advantages of the Separate-Reduction Approach . . . 16
  - 2.8.3 Implementation Considerations . . . 16

Notations The metric reads: $\mathrm{d}s^2 = a^2(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2)$, where the scale factor is $a(\tau) = -1/(H\tau)$. We set $H \equiv 1$ throughout this note.

1 Massless Case in de Sitter Space

For a massless scalar field in de Sitter space, the mode functions simplify significantly compared to the massive case. The equation of motion for a massless scalar field ϕ in de Sitter background

is:

$$\phi'' + 2\mathcal{H}\phi' - \nabla^2\phi = 0, \tag{1}$$

where $\mathcal{H} = a'/a = -1/\tau$ is the conformal Hubble parameter.

In Fourier space, the mode equation becomes:

$$u_k'' + \left(k^2 - \frac{2}{\tau^2}\right) u_k = 0. \tag{2}$$

The two independent solutions are:

$$u_k(\tau) = \frac{e^{-ik\tau}}{\sqrt{2k}}\left(1 - \frac{i}{k\tau}\right), \qquad u_k^*(\tau) = \frac{e^{ik\tau}}{\sqrt{2k}}\left(1 + \frac{i}{k\tau}\right). \tag{3}$$

These correspond to the Bunch-Davies vacuum.

For the massless case ($m = 0$, which corresponds to $\nu = 3/2$ in the massive notation), the Hankel functions simplify to elementary functions:

$$h_{3/2}^{(1)}(x) = -\sqrt{\frac{2}{\pi}} x^{-3/2} e^{ix}(1 + ix), \qquad h_{3/2}^{(2)}(x) = -\sqrt{\frac{2}{\pi}} x^{-3/2} e^{-ix}(1 - ix). \tag{4}$$

1.1 Feynman Rules for Massless Scalar Field

The Feynman rules for a massless scalar field in de Sitter space are:

1. **Propagators:** In the in-in (Schwinger-Keldysh) formalism, there are four bulk-to-bulk propagators corresponding to different time-orderings on the closed time path. We denote $u_k(\tau)$ as the positive-frequency solution and $u_k^*(\tau)$ as its complex conjugate, representing negative frequency. The four propagators are:

$$G_{++}(k;\tau_1,\tau_2) = u_k(\tau_1)u_k^*(\tau_2)\theta(\tau_1 - \tau_2) + u_k^*(\tau_1)u_k(\tau_2)\theta(\tau_2 - \tau_1) = \langle T\phi(\tau_1)\phi(\tau_2)\rangle, \tag{5}$$

$$G_{--}(k;\tau_1,\tau_2) = u_k^*(\tau_1)u_k(\tau_2)\theta(\tau_1 - \tau_2) + u_k(\tau_1)u_k^*(\tau_2)\theta(\tau_2 - \tau_1) = \langle \tilde{T}\phi(\tau_1)\phi(\tau_2)\rangle, \tag{6}$$

$$G_{+-}(k;\tau_1,\tau_2) = u_k(\tau_1)u_k^*(\tau_2) = \langle \phi(\tau_1)\phi(\tau_2)\rangle, \tag{7}$$

$$G_{-+}(k;\tau_1,\tau_2) = u_k^*(\tau_1)u_k(\tau_2) = \langle \phi(\tau_2)\phi(\tau_1)\rangle, \tag{8}$$

where T denotes time-ordering and $\tilde{T}$ anti-time-ordering. Note that $G_{--} = G_{++}^*$ and $G_{-+} = G_{+-}^*$. For the bubble diagram calculation, only G_{++} propagators appear when both vertices are on the $+$ branch of the time contour.

2. **Vertex:** Each vertex comes with a factor of $(-\tau)^n$ where n depends on the interaction. For a $\lambda\phi^4$ interaction, $n = 4$.

3. **Time integration:** Each vertex is integrated over conformal time τ from $-\infty$ to 0.

4. **Momentum conservation:** At each vertex, the sum of incoming momenta equals the sum of outgoing momenta.

1.2 One-Loop Two-Point Function for Massless Case

For the massless case, we can define a simpler integral family. Consider the one-loop two-point function (bubble diagram) for a massless scalar field:

$$\begin{aligned}\mathcal{G}[\{a_1,a_2\},\{b_1,b_2\}] \equiv \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-iP_1\tau_1 - iP_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} \\ \times\, u_q(-\tau_1)u_q^*(-\tau_2)u_{|\boldsymbol{q}+\boldsymbol{k}_s|}(-\tau_1)u^*_{|\boldsymbol{q}+\boldsymbol{k}_s|}(-\tau_2).\end{aligned} \tag{9}$$

Alternatively, we can write this in a more symmetric form using the mode functions explicitly:

$$\begin{aligned}\mathcal{G}[\{a_1,a_2\},\{b_1,b_2\}] = &\int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-iP_1\tau_1 - iP_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} \\ &\times \frac{e^{iq\tau_1}}{\sqrt{2q}}\left(1-\frac{i}{q\tau_1}\right)\frac{e^{-iq\tau_2}}{\sqrt{2q}}\left(1+\frac{i}{q\tau_2}\right) \\ &\times \frac{e^{i|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1}}{\sqrt{2|\boldsymbol{q}+\boldsymbol{k}_s|}}\left(1-\frac{i}{|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1}\right)\frac{e^{-i|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2}}{\sqrt{2|\boldsymbol{q}+\boldsymbol{k}_s|}}\left(1+\frac{i}{|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2}\right).\end{aligned} \tag{10}$$

This can be simplified to:

$$\begin{aligned}\mathcal{G}[\{a_1,a_2\},\{b_1,b_2\}] = &\frac{1}{4}\int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1+1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2+1}} \\ &\times e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1)\tau_1} e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2)\tau_2} \\ &\times \left(1-\frac{i}{q\tau_1}\right)\left(1+\frac{i}{q\tau_2}\right)\left(1-\frac{i}{|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1}\right)\left(1+\frac{i}{|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2}\right).\end{aligned} \tag{11}$$

1.3 Integral Family for IBP Analysis

For integration-by-parts (IBP) analysis, we need a general integral family that captures all possible structures arising from time and momentum derivatives. The simplified form below corresponds to a G_{+-}-type correlator without θ-function time-ordering, obtained by absorbing the mode-function products $u^{(1)}u^{(2)}$ from two G_{+-} propagators into exponential and denominator factors:

$$\begin{aligned}G[\{a_1,a_2\},\{b_1,b_2\}] \equiv \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} \\ \times\, e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1)\tau_1} e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2)\tau_2}.\end{aligned} \tag{12}$$

This simplified integral family captures the essential algebraic structure for IBP reduction while avoiding the complications of θ-functions. The full vertex++ case with time-ordering is treated in Section 2.

1.4 IBP Relations

The IBP relations for this family come from:

1. **Time derivatives:** $\int \partial_{\tau_i}[\cdots] = 0$
2. **Momentum derivatives:** $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ and $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i)\cdots] = 0$

The time IBP generates relations between integrals with different (a_i, b_{i+2}). Consider the derivative with respect to τ_1:

$$\begin{aligned} 0 =& \int_{-\infty}^{0} \partial_{\tau_1} \left[e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1)\tau_1} (-\tau_1)^{a_1} \right] \cdots \\ =& \int_{-\infty}^{0} \left[i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1) e^{iE_1\tau_1} (-\tau_1)^{a_1} - a_1 e^{iE_1\tau_1} (-\tau_1)^{a_1-1} \right] \cdots \\ =& i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1) G[\{a_1,a_2\},\{b_1,b_2\}] - a_1 G[\{a_1-1,a_2\},\{b_1,b_2\}], \end{aligned} \tag{13}$$

where $E_1 = q + |\boldsymbol{q} + \boldsymbol{k}_s| - P_1$. Note that the factor $(q + |\boldsymbol{q} + \boldsymbol{k}_s| - P_1)$ remains in the integrand as a momentum-dependent coefficient.

Similarly, for τ_2:

$$\begin{aligned} 0 =& \int_{-\infty}^{0} \partial_{\tau_2} \left[e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2)\tau_2} (-\tau_2)^{a_2} \right] \cdots \\ =& \int_{-\infty}^{0} \left[-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2) e^{-iE_2\tau_2} (-\tau_2)^{a_2} - a_2 e^{-iE_2\tau_2} (-\tau_2)^{a_2-1} \right] \cdots \\ =& -i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2) G[\{a_1,a_2\},\{b_1,b_2\}] - a_2 G[\{a_1,a_2-1\},\{b_1,b_2\}], \end{aligned} \tag{14}$$

where $E_2 = q + |\boldsymbol{q} + \boldsymbol{k}_s| + P_2$ and the minus sign comes from $\partial_\tau e^{-iE\tau} = -iEe^{-iE\tau}$.

Momentum IBP generates relations involving shifts in b_i and a_i due to the chain rule when differentiating $|\boldsymbol{q}|$ and $|\boldsymbol{k}_s + \boldsymbol{q}|$.

1.5 Momentum IBP Relations

Following the notation from the massive case, we denote:

$$x_1 \equiv q = |\boldsymbol{q}|, \qquad x_2 \equiv |\boldsymbol{q} + \boldsymbol{k}_s|. \tag{15}$$

The energies are:

$$E_1 = x_1 + x_2 - P_1, \qquad E_2 = x_1 + x_2 + P_2. \tag{16}$$

1.6 Momentum IBP Relations

The momentum IBP relations come from the identities $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ and $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i)\cdots] = 0$. To derive these relations systematically, we need the action of the derivative operators on the variables $x_1 = q$ and $x_2 = |\boldsymbol{q} + \boldsymbol{k}_s|$.

Define the derivative operators:

$$\mathcal{O}_1 \equiv \boldsymbol{q}^i \partial_{\boldsymbol{q}^i}, \tag{17}$$
$$\mathcal{O}_2 \equiv (\boldsymbol{q}^i + \boldsymbol{k}_s^i)\partial_{\boldsymbol{q}^i}. \tag{18}$$

These operators act on x_1 and x_2 as:

$$\mathcal{O}_1 x_1 = x_1, \tag{19}$$
$$\mathcal{O}_1 x_2 = \frac{1}{2} x_2^{-1}(x_1^2 + x_2^2 - k_s^2), \tag{20}$$
$$\mathcal{O}_2 x_1 = \frac{1}{2} x_1^{-1}(x_1^2 + x_2^2 - k_s^2), \tag{21}$$
$$\mathcal{O}_2 x_2 = x_2. \tag{22}$$

For the energy parameters $E_1 = x_1 + x_2 - P_1$ and $E_2 = x_1 + x_2 + P_2$, we have:

$$\mathcal{O}_1 E_1 = \mathcal{O}_1 E_2 = x_1 + \frac{1}{2} x_2^{-1}(x_1^2 + x_2^2 - k_s^2), \tag{23}$$
$$\mathcal{O}_2 E_1 = \mathcal{O}_2 E_2 = \frac{1}{2} x_1^{-1}(x_1^2 + x_2^2 - k_s^2) + x_2. \tag{24}$$

The full momentum IBP relations involve applying these derivative operators to all factors in the integrand and then integrating by parts. The resulting expressions are lengthy and will be derived and verified through code implementation. For now, we note that the IBP relations take the general form:

$$\begin{aligned} 0 = {} & D\, G[\{a_1, a_2\}, \{b_1, b_2\}] \\ & - \left[\frac{b_1}{x_1} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_1) G[\{a_1, a_2\}, \{b_1, b_2\}] \\ & - \left[\frac{b_2}{x_2} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_2) G[\{a_1, a_2\}, \{b_1, b_2\}], \end{aligned} \tag{25}$$

and a similar relation for $\mathcal{O}_2$, where D is the spacetime dimension.

The explicit form of these relations in terms of shifted G integrals will be provided after code verification, as the expressions are complex and best handled computationally.

1.7 Symmetry Relations Under Propagator Exchange

The bubble integral exhibits an important symmetry under the exchange of the two propagators. Consider the original bubble integral definition:

$$\begin{aligned} G[\{a_1, a_2\}, \{b_1, b_2\}] \equiv {} & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s + \boldsymbol{q}|^{b_2}} \\ & \times e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1)\tau_1} e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2)\tau_2}. \end{aligned} \tag{26}$$

Under the transformation $\boldsymbol{q} \to -\boldsymbol{q} - \boldsymbol{k}_s$, which exchanges the two propagators $|\boldsymbol{q}| \leftrightarrow |\boldsymbol{q} + \boldsymbol{k}_s|$, we have:

$$q = |\boldsymbol{q}| \to |-\boldsymbol{q} - \boldsymbol{k}_s| = |\boldsymbol{q} + \boldsymbol{k}_s| = x_2, \tag{27}$$

$$|\boldsymbol{q} + \boldsymbol{k}_s| \to |-\boldsymbol{q}| = |\boldsymbol{q}| = x_1. \tag{28}$$

The energy combinations remain invariant under this transformation:

$$q + |\boldsymbol{q} + \boldsymbol{k}_s| \to |\boldsymbol{q} + \boldsymbol{k}_s| + |\boldsymbol{q}| = x_2 + x_1, \tag{29}$$

so $E_1 = x_1 + x_2 - P_1$ and $E_2 = x_1 + x_2 + P_2$ are also invariant.

The integration measure $\mathrm{d}^3\boldsymbol{q}$ is invariant under this transformation. Therefore, the integral satisfies the following symmetry relation:

$$G[\{a_1, a_2\}, \{b_1, b_2\}] = G[\{a_1, a_2\}, \{b_2, b_1\}]. \tag{30}$$

This symmetry expresses the fact that the bubble diagram is symmetric under the exchange of the two internal propagators. In terms of the physical interpretation, this corresponds to the invariance of the one-loop two-point function under relabeling of the two internal lines in the bubble diagram.

1.8 Complete IBP System

The complete integration-by-parts system comprises four classes of relations, which together enable systematic reduction of any $G[\{a_1, a_2\}, \{b_1, b_2\}]$ integral.

1.8.1 Time IBP Relations

From vanishing surface terms $\int \partial_{\tau_i}[\cdots] = 0$:

$$i(q + |\boldsymbol{q} + \boldsymbol{k}_s| - P_1)G[\{a_1, a_2\}, \{b_1, b_2\}] - a_1 G[\{a_1 - 1, a_2\}, \{b_1, b_2\}] = 0, \tag{31}$$

$$-i(q + |\boldsymbol{q} + \boldsymbol{k}_s| + P_2)G[\{a_1, a_2\}, \{b_1, b_2\}] - a_2 G[\{a_1, a_2 - 1\}, \{b_1, b_2\}] = 0. \tag{32}$$

1.8.2 Momentum IBP Relations

From $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ and $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i) \cdots] = 0$:

$$\begin{aligned} 0 = {} & D\, G[\{a_1, a_2\}, \{b_1, b_2\}] \\ & - \left[\frac{b_1}{x_1} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_1)G[\{a_1, a_2\}, \{b_1, b_2\}] \\ & - \left[\frac{b_2}{x_2} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_2)G[\{a_1, a_2\}, \{b_1, b_2\}], \end{aligned} \tag{33}$$

and an analogous relation for $\mathcal{O}_2$, where $D = 3$ is the spacetime dimension, $x_1 = q$, $x_2 = |\boldsymbol{q} + \boldsymbol{k}_s|$, $E_1 = x_1 + x_2 - P_1$, $E_2 = x_1 + x_2 + P_2$, and the derivative actions are

$$\mathcal{O}_1 x_1 = x_1, \qquad \mathcal{O}_1 x_2 = \frac{1}{2} x_2^{-1}(x_1^2 + x_2^2 - k_s^2),$$

$$\mathcal{O}_2 x_1 = \frac{1}{2} x_1^{-1}(x_1^2 + x_2^2 - k_s^2), \qquad \mathcal{O}_2 x_2 = x_2.$$

1.8.3 Symmetry Relations

The integral family exhibits two discrete symmetries:

$$G[\{a_1, a_2\}, \{b_1, b_2\}] = G[\{a_1, a_2\}, \{b_2, b_1\}]. \tag{34}$$

2 Vertex++ IBP Relations

Section 1 presented IBP relations for the simplified G_{+-}-type bubble integral, which lacks θ-function time-ordering. For the physical vertex++ correlator containing θ-functions, the IBP relations require careful treatment of boundary terms arising from derivatives of step functions. This section derives the complete IBP system for the genuine vertex++ case, denoted as V_{++} to avoid confusion with propagators.

Strategy choice: Although the physical observable is the sum $\mathcal{V}_{++} = V_{++}^{(1)} + V_{++}^{(2)}$, we choose to derive IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$ separately rather than for their sum directly. The reason is that the time-derivative IBP relations for the sum would involve terms with opposite signs from the exponential factors ($iE_1^{(1)}$ for $V_{++}^{(1)}$ and $-iE_1^{(2)}$ for $V_{++}^{(2)}$), preventing systematic pairing in the reduction process. By handling each time-ordering independently and allowing for relabeling of external energies $P_1 \leftrightarrow -P_2$ when needed, we maintain a consistent algebraic structure that mirrors the G_{+-} case, with boundary terms $\pm H$ as the only additional complexity. This separate-treatment approach simplifies the reduction procedure while retaining the option to reconstruct the sum $\mathcal{V}_{++}$ after reduction.

2.1 Definition of Vertex++ Integrals

The vertex++ bubble integral naturally decomposes into two θ-ordered pieces:

$$\mathcal{V}_{++} = V_{++}^{(1)} + V_{++}^{(2)}, \tag{35}$$

where $V_{++}^{(1)}$ contains $\theta(\tau_1 - \tau_2)$ and $V_{++}^{(2)}$ contains $\theta(\tau_2 - \tau_1)$. Writing out the explicit form after absorbing the mode-function products into exponential factors as in Section 1.3:

$$\begin{aligned} V_{++}^{(1)}[\{a_1, a_2\}, \{b_1, b_2\}] = \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s + \boldsymbol{q}|^{b_2}} \\ \times e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_1)\tau_1} e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_2)\tau_2} \cdot \theta(\tau_1 - \tau_2), \end{aligned} \tag{36}$$

$$\begin{aligned} V_{++}^{(2)}[\{a_1, a_2\}, \{b_1, b_2\}] = \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s + \boldsymbol{q}|^{b_2}} \\ \times e^{-i(q+|\boldsymbol{q}+\boldsymbol{k}_s|+P_1)\tau_1} e^{i(q+|\boldsymbol{q}+\boldsymbol{k}_s|-P_2)\tau_2} \cdot \theta(\tau_2 - \tau_1). \end{aligned} \tag{37}$$

The two definitions differ in two respects:

1. The exponential factors: $V_{++}^{(1)}$ has $e^{iE_1^{(1)}\tau_1} e^{-iE_2^{(1)}\tau_2}$ while $V_{++}^{(2)}$ has $e^{-iE_1^{(2)}\tau_1} e^{iE_2^{(2)}\tau_2}$, where $E_1^{(1)} = q + |\boldsymbol{q}+\boldsymbol{k}_s| - P_1$, $E_2^{(1)} = q + |\boldsymbol{q}+\boldsymbol{k}_s| + P_2$, $E_1^{(2)} = q + |\boldsymbol{q}+\boldsymbol{k}_s| + P_1$, and $E_2^{(2)} = q + |\boldsymbol{q}+\boldsymbol{k}_s| - P_2$.

2. The θ-functions: $\theta(\tau_1 - \tau_2)$ vs $\theta(\tau_2 - \tau_1)$.

Note that $V_{++}^{(1)}$ corresponds to the product $u_q(\tau_1)u_q^*(\tau_2)u_{|\boldsymbol{q}+\boldsymbol{k}_s|}(\tau_1)u_{|\boldsymbol{q}+\boldsymbol{k}_s|}^*(\tau_2)$ while $V_{++}^{(2)}$ corresponds to the conjugate product $u_q^*(\tau_1)u_q(\tau_2)u_{|\boldsymbol{q}+\boldsymbol{k}_s|}^*(\tau_1)u_{|\boldsymbol{q}+\boldsymbol{k}_s|}(\tau_2)$. For IBP analysis, we keep the θ-functions outside the integration; they affect only the time-derivative IBP relations, as we now discuss.

2.2 Boundary Terms from Time Derivatives

When applying a time derivative ∂_{τ_i} to $V_{++}^{(1)}$ or $V_{++}^{(2)}$, the derivative acts both on the smooth exponential factors and on the θ-function. The action on the smooth part generates IBP relations similar to those in Section 1.4, while the derivative of the θ-function generates $\delta(\tau_1 - \tau_2)$ boundary terms.

Defining the boundary integral family

$$H[c, \{b_1, b_2\}] \equiv \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-i(P_1+P_2)\tau}(-\tau)^c \frac{1}{x_1^{b_1} x_2^{b_2}}, \tag{38}$$

which corresponds to collapsing the two time vertices to $\tau_1 = \tau_2 \equiv \tau$. This represents the contribution from $\delta(\tau_1 - \tau_2)$ where the exponential factors simplify to $e^{-i(P_1+P_2)\tau}$.

For $V_{++}^{(1)}$, applying ∂_{τ_1} gives:

$$\begin{aligned}
&\int \partial_{\tau_1} \left[e^{iE_1^{(1)}\tau_1}(-\tau_1)^{a_1}\theta(\tau_1 - \tau_2)\cdots \right] \\
&= \int \Big[iE_1^{(1)} e^{iE_1^{(1)}\tau_1}(-\tau_1)^{a_1}\theta(\tau_1 - \tau_2) - a_1 e^{iE_1^{(1)}\tau_1}(-\tau_1)^{a_1-1}\theta(\tau_1 - \tau_2) \\
&\quad + e^{iE_1\tau_1}(-\tau_1)^{a_1}\delta(\tau_1 - \tau_2)\Big]\cdots
\end{aligned} \tag{39}$$

The δ-function term collapses the time integration to $\tau_1 = \tau_2 \equiv \tau$, producing $H[a_1 + a_2, \{b_1, b_2\}]$. Thus we obtain:

$$iE_1^{(1)} V_{++}^{(1)}[\{a_1, a_2\}, \{b_1, b_2\}] - a_1 V_{++}^{(1)}[\{a_1 - 1, a_2\}, \{b_1, b_2\}] + H[a_1 + a_2, \{b_1, b_2\}] = 0. \tag{40}$$

For $V_{++}^{(2)}$, applying ∂_{τ_1} to $e^{-iE_1^{(2)}\tau_1}(-\tau_1)^{a_1}\theta(\tau_2 - \tau_1)$ gives:

$$\begin{aligned}
&\int \partial_{\tau_1} \left[e^{-iE_2\tau_1}(-\tau_1)^{a_1}\theta(\tau_2 - \tau_1)\cdots \right] \\
&= \int \Big[-iE_1^{(2)} e^{-iE_1^{(2)}\tau_1}(-\tau_1)^{a_1}\theta(\tau_2 - \tau_1) - a_1 e^{-iE_1^{(2)}\tau_1}(-\tau_1)^{a_1-1}\theta(\tau_2 - \tau_1) \\
&\quad - e^{-iE_1^{(2)}\tau_1}(-\tau_1)^{a_1}\delta(\tau_1 - \tau_2)\Big]\cdots
\end{aligned} \tag{41}$$

Note the minus sign from $\partial_{\tau_1}\theta(\tau_2 - \tau_1) = -\delta(\tau_1 - \tau_2)$. The δ-function term again produces $H[a_1 + a_2, \{b_1, b_2\}]$, yielding:

$$-iE_1^{(2)} V_{++}^{(2)}[\{a_1, a_2\}, \{b_1, b_2\}] - a_1 V_{++}^{(2)}[\{a_1 - 1, a_2\}, \{b_1, b_2\}] - H[a_1 + a_2, \{b_1, b_2\}] = 0. \tag{42}$$

Similar relations can be derived from ∂_{τ_2}, but they provide redundant information due to the symmetry between the two time variables.

Comparing with the G_{+-} time IBP (Eqs. (1.6)–(1.7)), the only new ingredients are the boundary terms $\pm H$, which originate from derivatives of the θ-functions. The coefficients $iE_1^{(1)}$ and $-iE_1^{(2)}$ appear naturally from differentiating the exponential factors.

2.3 Reduction of Boundary Integrals H

The boundary integrals $H[c, \{b_1, b_2\}]$ defined in Eq. (2.2) are single-time integrals that appear in the time IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$. To complete the reduction system, we need IBP relations for H itself. These integrals are simpler than the double-time V_{++} integrals because they contain only one time variable τ.

2.3.1 Time IBP for H

Applying a time derivative to the integrand of H gives:

$$\partial_\tau \left[e^{-i(P_1+P_2)\tau}(-\tau)^c\right] = -i(P_1+P_2)e^{-i(P_1+P_2)\tau}(-\tau)^c - ce^{-i(P_1+P_2)\tau}(-\tau)^{c-1}. \tag{43}$$

Assuming the surface terms vanish at $\tau = -\infty$ and $\tau = 0$ (which holds for appropriate analytic continuations of $P_1 + P_2$), we obtain the time IBP relation:

$$-i(P_1+P_2)H[c, \{b_1, b_2\}] - cH[c-1, \{b_1, b_2\}] = 0. \tag{44}$$

This relation reduces the time exponent c at the cost of introducing a factor of $1/(P_1 + P_2)$.

2.3.2 Momentum IBP for H

Momentum derivatives act only on the denominator $1/(x_1^{b_1} x_2^{b_2})$. Applying the operators $\mathcal{O}_1 = \boldsymbol{q}^i \partial_{\boldsymbol{q}^i}$ and $\mathcal{O}_2 = (\boldsymbol{q}^i + \boldsymbol{k}_s^i)\partial_{\boldsymbol{q}^i}$ to the integrand yields:

$$\mathcal{O}_1 \frac{1}{x_1^{b_1} x_2^{b_2}} = -\frac{b_1}{x_1^{b_1} x_2^{b_2}} - \frac{b_2}{x_2^{b_2+1}} \mathcal{O}_1 x_2, \tag{45}$$

$$\mathcal{O}_2 \frac{1}{x_1^{b_1} x_2^{b_2}} = -\frac{b_1}{x_1^{b_1+1}} \mathcal{O}_2 x_1 - \frac{b_2}{x_2^{b_2} x_2^{b_2}}. \tag{46}$$

Using $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ and $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i) \cdots] = 0$, and noting that the exponential factor is independent of $\boldsymbol{q}$, we obtain the momentum IBP relations:

$$0 = D\, H[c, \{b_1, b_2\}] - \frac{b_1}{x_1}(\mathcal{O}_1 x_1) H[c, \{b_1, b_2\}] - \frac{b_2}{x_2}(\mathcal{O}_1 x_2) H[c, \{b_1, b_2\}], \tag{47}$$

$$0 = D\, H[c, \{b_1, b_2\}] - \frac{b_1}{x_1}(\mathcal{O}_2 x_1) H[c, \{b_1, b_2\}] - \frac{b_2}{x_2}(\mathcal{O}_2 x_2) H[c, \{b_1, b_2\}], \tag{48}$$

where $D = 3$ is the spatial dimension and the derivative actions $\mathcal{O}_i x_j$ are the same as in Eqs. (1.8)–(1.9). Since H contains no factors of τ_1, τ_2, the coefficients $-i\tau_1 + i\tau_2$ that appear in the G_{+-} momentum IBP are absent here.

2.3.3 Symmetry Relations

The H integrals inherit the symmetry under exchange of the two propagators from the original bubble diagram:

$$H[c, \{b_1, b_2\}] = H[c, \{b_2, b_1\}]. \tag{49}$$

This follows from the invariance of $x_1 x_2$ under $x_1 \leftrightarrow x_2$ and the invariance of the measure $\mathrm{d}^3\boldsymbol{q}$.

2.3.4 Reduction Strategy

The IBP system for H is simpler than that for V_{++} because:

1. Only one time variable appears, so time IBP involves only a single relation (Eq. 44).
2. The momentum IBP relations (Eqs. 47–48) lack the τ-dependent terms, making them algebraic in the b_i indices.
3. The symmetry (Eq. 49) reduces the number of independent integrals by a factor of two when $b_1 \neq b_2$.

In practice, one can reduce any $H[c, \{b_1, b_2\}]$ to a set of master integrals (e.g., $H[0, \{0, 0\}]$, $H[0, \{1, 0\}]$, etc.) using the above relations. Since H integrals appear only as inhomogeneous terms in the V_{++} IBP system, their reduction can be performed once and the results substituted into the V_{++} reduction.

2.4 Momentum IBP Relations

Since momentum derivatives commute with the θ-functions, the momentum IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$ are identical to those for the G_{+-} family $G[\{a_i\}, \{b_i\}]$ derived in Section 1.6. One simply replaces G by $V_{++}^{(1)}$ or $V_{++}^{(2)}$ in Eqs. (1.10)–(1.11). Explicitly,

$$\begin{aligned} 0 = {} & D\, V_{++}^{(1)}[\{a_i\}, \{b_i\}] \\ & - \left[\frac{b_1}{x_1} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_1) V_{++}^{(1)}[\{a_i\}, \{b_i\}] \\ & - \left[\frac{b_2}{x_2} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_2) V_{++}^{(1)}[\{a_i\}, \{b_i\}], \end{aligned} \tag{50}$$

and analogously for $V_{++}^{(2)}$ (with the same coefficients, as discussed below) and for the operator $\mathcal{O}_2$. The derivative actions $\mathcal{O}_1 x_1$, $\mathcal{O}_1 x_2$, etc., are exactly as in Section 1.6. Remarkably, the momentum IBP relations for both $V_{++}^{(1)}$ and $V_{++}^{(2)}$ are identical to those for G_{+-} (Eq. (1.10)), despite the different exponential factors.

2.5 Algebraic Relations and Symmetries

The discrete symmetries are inherited from the structure of the integrands. Both $V_{++}^{(1)}$ and $V_{++}^{(2)}$ are invariant under exchange of the two propagators $x_1 \leftrightarrow x_2$, which corresponds to the transformation $\boldsymbol{q} \to -\boldsymbol{q} - \boldsymbol{k}_s$:

$$V_{++}^{(1)}[\{a_1, a_2\}, \{b_1, b_2\}] = V_{++}^{(1)}[\{a_1, a_2\}, \{b_2, b_1\}], \tag{51}$$

$$V_{++}^{(2)}[\{a_1, a_2\}, \{b_1, b_2\}] = V_{++}^{(2)}[\{a_1, a_2\}, \{b_2, b_1\}]. \tag{52}$$

This symmetry follows from the invariance of $x_1 + x_2$ under $x_1 \leftrightarrow x_2$ and the invariance of the measure $\mathrm{d}^3\boldsymbol{q}$.

When $a_1 = a_2$, there is a relation between the two time-orderings via simultaneous exchange of time variables and a sign flip of the external energies. Performing the transformations $\tau_1 \leftrightarrow \tau_2$ and $P_1 \leftrightarrow -P_2$ maps $V_{++}^{(2)}$ to $V_{++}^{(1)}$:

$$V_{++}^{(2)}[\{a,a\},\{b_1,b_2\};P_1,P_2] = V_{++}^{(1)}[\{a,a\},\{b_1,b_2\};-P_2,-P_1], \qquad (a_1 = a_2 = a). \tag{53}$$

For general $a_1 \neq a_2$, no simple relation connects $V_{++}^{(1)}$ and $V_{++}^{(2)}$ due to the different dependence on τ_1 and τ_2 in the time exponents.

2.6 Complete IBP System for V_{++}

Collecting the results above, the IBP system for the V_{++} bubble integrals is obtained from the G_{+-} system of Section 1 with the sole addition of boundary terms coming from derivatives of the θ-functions:

1. **Time IBP relations:** Eqs. (2.3)–(2.4), which are the G_{+-} time IBP relations plus the boundary terms $+H$ for $V_{++}^{(1)}$ and $-H$ for $V_{++}^{(2)}$.
2. **Momentum IBP relations:** Identical to Eqs. (1.10)–(1.11) with G replaced by $V_{++}^{(1)}$ or $V_{++}^{(2)}$.
3. **Symmetry relations:** Eqs. (51)–(53) above.

The only genuinely new ingredients are the boundary integrals $H[c,\{b_i\}]$ defined in Eq. (2.2), which couple the two time-ordered sectors $V_{++}^{(1)}$ and $V_{++}^{(2)}$.

2.7 Relation to Chapter 1 IBP and Separate Reduction Strategy

A crucial observation is that the momentum IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$ are *identical* to those for the G_{+-} family derived in Section 1. Specifically, comparing (50) with Eq. (1.10), we see:

$$\text{Momentum IBP for } V_{++}^{(1)}: \quad 0 = D\,V_{++}^{(1)} - \left[\frac{b_1}{x_1} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_1)V_{++}^{(1)} - \left[\frac{b_2}{x_2} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_2)V_{++}^{(1)}, \tag{54}$$

$$\text{Momentum IBP for } G_{+-}: \quad 0 = D\,G - \left[\frac{b_1}{x_1} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_1)G - \left[\frac{b_2}{x_2} - i\tau_1 + i\tau_2\right](\mathcal{O}_1 x_2)G. \tag{55}$$

The mathematical structure is identical, with only the integral symbol ($V_{++}^{(1)}$ vs G) differing. This means:

- **Direct application of Chapter 1 results:** Any algebraic manipulation, reduction formula, or master integral basis derived for G_{+-} in Section 1 applies directly to $V_{++}^{(1)}$ and $V_{++}^{(2)}$ for the momentum-space part of the reduction.

- **Time IBP similarity:** The time IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$ differ from those of G_{+-} only by the boundary terms $\pm H$. The core differential operators are identical when written in terms of the appropriate energies $E_1^{(1)}$ and $E_1^{(2)}$.

- **Symmetry inheritance:** The symmetry $V_{++}^{(1)}[\{a_1, a_2\}, \{b_1, b_2\}] = V_{++}^{(1)}[\{a_1, a_2\}, \{b_2, b_1\}]$ is inherited directly from G_{+-}.

2.7.1 Why Separate Reduction is Preferred

Although the boundary terms $\pm H$ would cancel if we added the time IBP relations for $V_{++}^{(1)}$ and $V_{++}^{(2)}$, this cancellation comes at a cost: the exponential factors in the two time-orderings have opposite signs ($iE_1^{(1)}$ for $V_{++}^{(1)}$ and $-iE_1^{(2)}$ for $V_{++}^{(2)}$). Consequently, the coefficients in the time IBP relations do not pair systematically when considering the sum $\mathcal{V}_{++} = V_{++}^{(1)} + V_{++}^{(2)}$. This lack of pairing complicates the reduction process if one attempts to work directly with the sum.

Instead, we adopt a strategy of reducing $V_{++}^{(1)}$ and $V_{++}^{(2)}$ separately, treating each as an independent integral family with its own IBP system. The two families are coupled only through the common boundary integrals H, which appear with opposite signs. This separate-treatment approach maintains the algebraic simplicity of the G_{+-} reduction while cleanly isolating the boundary effects.

2.7.2 Energy Relabeling as a Simplification Tool

An important simplification arises from the symmetry relation (53), which shows that $V_{++}^{(2)}$ with external energies (P_1, P_2) equals $V_{++}^{(1)}$ with energies $(-P_2, -P_1)$ when $a_1 = a_2$. Even for general $a_1 \neq a_2$, one can view $V_{++}^{(2)}$ as essentially the same mathematical object as $V_{++}^{(1)}$ but with a different assignment of energies to the time variables. This perspective allows us to:

1. Focus reduction efforts primarily on the $V_{++}^{(1)}$ family, using the results for $V_{++}^{(2)}$ via energy relabeling when $a_1 = a_2$.

2. Maintain a unified reduction procedure that handles both time-orderings through parameter substitution rather than separate derivations.

3. Exploit the fact that the momentum IBP relations are completely independent of the energy assignments P_1, P_2.

The boundary integrals H remain as the only genuinely new structures that cannot be mapped to the G_{+-} system. Their reduction, discussed in Section 2.3, is relatively simple due to their single-time nature.

2.7.3 Consequences for the Reduction Procedure

The separate-reduction strategy leads to the following workflow:

1. **Momentum reduction:** Apply all momentum IBP reductions from Section 1 directly to $V_{++}^{(1)}$ (and similarly to $V_{++}^{(2)}$ if needed), exactly as for G_{+-} integrals.

2. **Time reduction with boundary terms:** Use the time IBP relations (2.3)–(2.4) which include the $\pm H$ boundary terms. These relations have the same algebraic form as the G_{+-} time IBP, with the energies $E_1^{(1)}$ or $E_1^{(2)}$ appearing as coefficients.

3. **Boundary integral reduction:** Reduce the H integrals using their own IBP system (Section 2.3).

4. **Reconstruction of the sum:** After reducing $V_{++}^{(1)}$ and $V_{++}^{(2)}$ to master integrals, add them to obtain the physical observable $\mathcal{V}_{++}$. For $a_1 = a_2$, use the relabeling symmetry to obtain $V_{++}^{(2)}$ from $V_{++}^{(1)}$ without separate reduction.

This approach preserves the systematic structure of the IBP reduction while cleanly handling the boundary terms that distinguish the vertex++ case from the simpler G_{+-} case.

2.8 Practical Reduction Strategy

Based on the separate-reduction approach outlined in Section 2.7, the reduction of vertex++ bubble integrals follows a systematic procedure that treats $V_{++}^{(1)}$ and $V_{++}^{(2)}$ as independent integral families coupled through boundary terms. The strategy leverages the structural similarities with the G_{+-} case while cleanly handling the additional complexity introduced by θ-functions.

2.8.1 Reduction Procedure for Separate Time-Orderings

The recommended workflow for reducing vertex++ integrals is:

1. **Momentum reduction:** Apply all momentum IBP reductions from Section 1 directly to $V_{++}^{(1)}$ (and similarly to $V_{++}^{(2)}$ if needed), treating them exactly like G_{+-} integrals. The momentum IBP relations are identical to Eqs. (1.10)–(1.11).

2. **Time reduction with boundary terms:** Use the time IBP relations (2.3)–(2.4), which are the G_{+-} time IBP plus $\pm H$ boundary terms. For $V_{++}^{(1)}$, the coefficient is $iE_1^{(1)} = i(q + |\boldsymbol{q} + \boldsymbol{k}_s| - P_1)$; for $V_{++}^{(2)}$, it is $-iE_1^{(2)} = -i(q + |\boldsymbol{q} + \boldsymbol{k}_s| + P_1)$.

3. **Boundary integral reduction:** Reduce the H integrals using their own IBP system (Section 2.3). These single-time integrals are simpler and can be reduced to a small set of master integrals.

4. **Symmetry exploitation:** For $a_1 = a_2$, use the relabeling symmetry (53) to obtain $V_{++}^{(2)}$ from $V_{++}^{(1)}$ without separate reduction:

$$V_{++}^{(2)}[\{a, a\}, \{b_1, b_2\}; P_1, P_2] = V_{++}^{(1)}[\{a, a\}, \{b_1, b_2\}; -P_2, -P_1].$$

   This avoids duplicate work and highlights the essential unity of the two time-orderings.

5. **Reconstruction of the physical observable:** After reducing both $V_{++}^{(1)}$ and $V_{++}^{(2)}$ to master integrals, sum them to obtain the complete vertex++ correlator:

$$\mathcal{V}_{++}[\{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = V_{++}^{(1)}[\{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] + V_{++}^{(2)}[\{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2].$$

2.8.2 Key Advantages of the Separate-Reduction Approach

- **Systematic pairing:** By keeping $V_{++}^{(1)}$ and $V_{++}^{(2)}$ separate, the coefficients in the time IBP relations maintain consistent signs ($iE_1^{(1)}$ and $-iE_1^{(2)}$), allowing for orderly reduction without sign cancellations that would complicate the algebraic structure.

- **Clean boundary isolation:** The H integrals appear explicitly with opposite signs in the two time-orderings, making their role transparent. This clarity is lost if one attempts to reduce the sum $\mathcal{V}_{++}$ directly.

- **Flexibility:** The approach yields explicit expressions for individual time-orderings when needed, while still providing the sum $\mathcal{V}_{++}$ for physical observables.

- **Computational efficiency:** The momentum reduction reuses all results from Chapter 1, and the time reduction follows the same pattern as G_{+-} except for the addition of H terms. The H integrals themselves are simpler due to their single-time nature.

- **Relabeling symmetry:** The ability to map $V_{++}^{(2)}$ to $V_{++}^{(1)}$ via energy relabeling (when $a_1 = a_2$) reduces the effective number of independent integrals by nearly half.

2.8.3 Implementation Considerations

In practical implementations (e.g., using IBP reduction software), one can:

1. Define two integral families: one for $V_{++}^{(1)}$ and one for $V_{++}^{(2)}$, with the appropriate energy parameters $E_1^{(1)}$ and $E_1^{(2)}$ in the time IBP relations.

2. Include the H integrals as auxiliary masters that couple the two families through the boundary terms $\pm H$.

3. For $a_1 = a_2$, exploit the relabeling symmetry to reduce the computational cost.

4. After reduction, combine the master integrals for $V_{++}^{(1)}$ and $V_{++}^{(2)}$ to obtain the final result for $\mathcal{V}_{++}$.

This strategy maintains the mathematical rigor of the IBP procedure while providing a clear path from the defining integrals to reduced expressions suitable for numerical evaluation or analytic simplification.